

Vertical Motion

$$u = 0 \text{ ms}^{-1}$$

Take the downward direction as positive.

time	displacement		velocity
0	0		0
0.2 s	0.2 m		2 m s^{-1}
0.4 s	0.8 m		4 m s^{-1}
0.6 s	1.8 m		6 m s^{-1}
0.8 s	3.2 m		8 m s^{-1}
1 s	5 m		10 m s^{-1}

Take the **downward** direction as **positive**

When an object moves under the influence of gravity only, it always **accelerates downward at 9.81 m s^{-2}** .

$$a \text{ (acceleration)} = g \text{ (gravity)}$$

$$a = 9.81 \text{ ms}^{-2}$$

1. How will the acceleration value obtained affected if you release the object from a high position, and double the mass of the fall object?

Higher position

Double mass

- | | | |
|----|-----------|-----------|
| A. | increases | increases |
| B. | increases | no change |
| C. | no change | decreases |
| D. | no change | no change |

Acceleration of an object with **vertical motion** is always **equal to the gravity**.

D

2. Find the velocity, in ms^{-1} , of an object falling freely under gravity for a distance of 100 m.

- | | |
|---------|---------|
| A. 10 | B. 22.4 |
| C. 44.3 | D. 500 |

$$v^2 = 2(9.81)(100)$$

$$v = 44.29 \text{ ms}^{-1}$$

C

3. A small ball is projected from the top of a building, at an initial speed of 20 ms^{-1} . After 7.5 seconds, it reaches the ground. How tall is this building?

- A. 425.91 m
B. 281.25 m
C. 171.25 m
D. 131.25 m

$$S = 20(7.5) + \frac{1}{2}(9.81)7.5^2 = 429.5 \text{ m}$$

A

4. An object is falling from rest with an acceleration of 9.81 ms^{-2} . Which of the following statements is correct?

- A. It falls 9.81 m every second
B. The velocity of the object after 2 s is 9.81 ms^{-1}
C. Its displacement increases uniformly with time
D. Its velocity increases uniformly with time

An object falling from rest with an acceleration of 9.81 m s^{-2} means that its velocity increases 9.81 m s^{-1} every second, and so its velocity increases uniformly with time. The velocity of the object after 2 s is $9.81 + 9.81 = 19.6 \text{ m s}^{-1}$.

D

5. A stone is dropped from rest from the roof of a tall building. Find the distance travelled in the **THIRD SECOND**.

- A. 20 m
B. 25 m
C. 40 m
D. 45 m

$$u = 9.81(2) = 19.62 \text{ ms}^{-1} \quad v = (9.81)(3) = 29.43 \text{ ms}^{-1}$$

$$29.43^2 = 19.62^2 + 2(9.81)(s) \quad s = 24.525 \text{ ms}^{-1}$$

B

6. Neglecting air resistance, what is the distance travelled by a free body within the **fourth second** after it falls from rest?

- A. 34 m
B. 44 m
C. 54 m
D. 64 m

Total distance travelled in 4 seconds

$$= s_4 = ut + \frac{1}{2}at^2 = \frac{1}{2} \times 9.81 \times 4^2 = 78.48 \text{ m}$$

Total distance travelled in 3 seconds

$$= s_3 = \frac{1}{2} \times 9.81 \times 3^2 = 44.15 \text{ m}$$

$\therefore$ Distance travelled within the fourth second

$$= s_4 - s_3 = 34 \text{ m}$$

A

7. A stone falls freely from rest under gravity. If its speed is x at the end of the 1st second, what is its speed at the end of the 2nd second?

- A. x
B. $2x$
C. $3x$
D. $4x$

Acceleration = x
Speed after 2nd second = $2x$

B

8. A stone falls freely from rest under gravity. If it travels a distance y in the 1st second, what is the distance it travels in the 2nd second?

- A. y
B. $2y$
C. $3y$
D. $4y$

$$y = \frac{1}{2}(g) \quad \text{Distance travelled after 2nd second} = \frac{1}{2}(g)2^2 - \frac{1}{2}(g) = 1.5g$$

$$= 3y$$

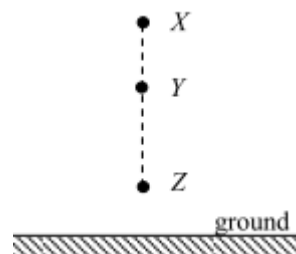
C

9. A particle is released from rest at X as shown. It takes time t_1 to fall from X to Y and time t_2 to fall from Y to Z. If $XY : YZ = 9 : 16$, find $t_1 : t_2$. Neglect air resistance.

- A. 2 : 3
B. 3 : 4
C. 4 : 3
D. 3 : 2

By $s = \frac{1}{2}gt^2$, $XY = \frac{1}{2}gt_1^2$ and $XZ = \frac{1}{2}g(t_1 + t_2)^2$

$$\frac{t_1^2}{(t_1 + t_2)^2} = \frac{XY}{XZ} = \frac{XY}{XY + YZ} = \frac{9}{9 + 16} = \frac{9}{25} \Rightarrow \frac{t_1}{t_1 + t_2} = \frac{3}{5} \Rightarrow \frac{t_1}{t_2} = \frac{3}{2}$$



D

10. A stone falls from rest from point X on a cliff. The time it takes t fall from X to Y is T and from Y and Z is $3T$. Find $XY : YZ$.

- A. 1 : 3
B. 1 : 9
C. 1 : 15
D. 1 : 16

Take downward as positive. Consider motion from X.
 $u = 0$, $a = +9.81 \text{ m s}^{-2}$

$$s = ut + \frac{1}{2}at^2 = 4.905t^2 \propto t^2$$

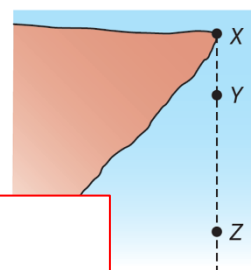
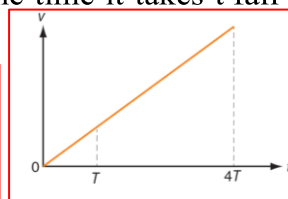
$$\therefore \frac{s_1}{s_2} = \frac{t_1^2}{t_2^2}$$

$$\frac{XY}{XZ} = \frac{T^2}{(T + 3T)^2} = \frac{1}{16}$$

If $XY = d$, then $XZ = 16d$.

$$YZ = 16d - d = 15d$$

$$XY : YZ = 1 : 15$$



$$\frac{XY}{XZ} = \frac{\text{area under graph from 0 to } T}{\text{area under graph from 0 to } 4T} = \frac{T^2}{(4T)^2} = \frac{1}{16}$$

If $XY = d$, then $XZ = 16d$.

$$YZ = 16d - d = 15d$$

$$XY : YZ = 1 : 15$$

$\therefore$ The answer is C.



C

The motion of a ball projected upwards

The time to reach
the highest point
= 2 s

time displacement velocity

2 s +20.0 m 0

0.8 s +12.8 m +12 m s⁻¹

0 0 +20 m s⁻¹

Take the upward direction be positive.

time displacement velocity

The highest point, $v = 0$

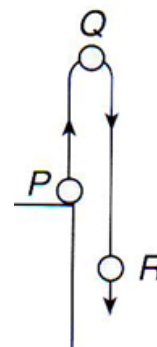
3.2 s +12.8 m -12 m s⁻¹

4 s 0 -20 m s⁻¹

Throughout the entire motion,
acceleration is always pointing
vertically downward of 9.81ms^{-2}

11. A piece of stone is thrown from the top of a tall building and its path is shown. Determine the direction of acceleration at different positions, P , Q and R .

	Position P	Position Q	Position R
A.	upwards	upwards	downwards
B.	upwards	no acceleration	upwards
C.	upwards	downwards	downwards
D.	downwards	downwards	downwards



D

12. Which of the following description about speed, velocity and acceleration is **incorrect**?
- A Velocity and acceleration are vectors.
- B An object can have zero acceleration but non-zero speed.
- C If an object has non-zero acceleration, its velocity must not be zero at all times.
- D If an object has zero acceleration, its average speed is equal to the instantaneous speed.

C

13. A ball is projected vertically upwards. After reaching the highest position Y , the ball returns to its original position. Which of the following statements are incorrect?

- (1) The direction of its velocity is the same as that of its acceleration throughout the motion
- (2) Its velocity just after being thrown is the same as that just before reaching its original position

D

- (3) At Y , both of its velocity and acceleration are zero

A. (1) and (2) only

B. (1) and (3) only

C. (2) and (3) only

D. (1), (2) and (3)

(3) * At Y , as the ball is changing its direction of motion, its instantaneous velocity is zero. However, its acceleration is always equal to 9.81m s^{-2} (= acceleration due to gravity).

(1) * During the upward flight, its velocity is upward while its acceleration is downward (because of acceleration due to gravity) for the whole period of flight.

(2) * By symmetry, the **magnitude** of its velocity just after being thrown is the same as that just before reaching its original position. However, their **directions** are opposite, and hence they are **not** equal.

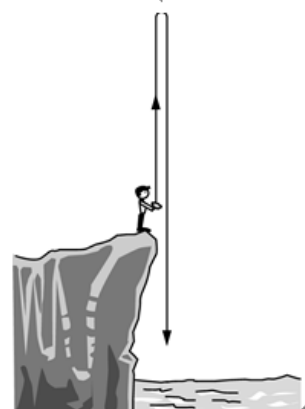
14. In the following figure, Tommy stands at a cliff and throws a stone vertically upwards at 15 m s^{-1} . The stone rises to the highest point and then falls into the sea. If Tommy is 50 m above the sea, what is the time interval for the stone moving in air? Take $g = 9.81 \text{ m s}^{-2}$.

- A 4 s
B 4.5 s
C 5.07 s
D 5.7 s

$$S = ut + \frac{1}{2}at^2 \quad \text{Take upward be positive}$$

$$-50 = 15t - \frac{1}{2}(9.81)t^2$$

$$t = -2.01 \text{ s (rejected)} \quad \text{or} \quad t = 5.07 \text{ s}$$



C

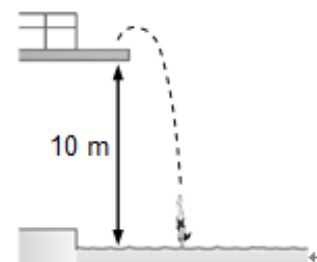
15. At $t = 0$, a diver jumps into a very deep pool from a height of 10 m with an initial upward velocity 0.8 m s^{-1} . Take $g = 9.81 \text{ m s}^{-2}$ and neglect air resistance. Find the time taken for the diver to reach the water.

- A 1.35 s
B 1.51 s
C 2.07 s
D 3.57 s

$$s = ut + \frac{1}{2}at^2$$

$$-10 = 0.8 \times t + \frac{1}{2} \times (-9.81) \times t^2$$

$$t = 1.51 \text{ s} \quad \text{or} \quad -1.35 \text{ s (rejected)}$$



B

16. An helicopter flying vertically upward with a uniform velocity of 3.6 km h^{-1} releases an object of mass 2 kg when it is at a height of 966 m . Neglecting air resistance, the object will land after

- A 13.9 s
B 14.1 s
C 14.3 s
D 14.5 s

Taking 'upward' direction as positive,

$$u = 3.6 \times \frac{1000}{3600} = 1 \text{ m s}^{-1},$$

$$s = -966 \text{ m},$$

$$a = g = -9.81 \text{ m s}^{-2}$$

$$\text{By } s = ut + \frac{1}{2}at^2$$

$$-966 = 1 \times t + \frac{1}{2}at^2 = t - 4.905t^2$$

$$\therefore 4.905t^2 - t - 966 = 0$$

$$t = 14.1 \text{ s} \quad \text{or} \quad t = -13.9 \text{ s (rejected)}$$

B

17. A dolphin jumps up to touch a ball which is hung at 6.8 m above the water surface. What is the minimum possible speed of the dolphin when it just leaves the water surface?

- A 4.7 m s^{-1}
B 7.1 m s^{-1}
C 8.2 m s^{-1}
D 11.6 m s^{-1}

$$v^2 = u^2 + 2as$$

$$0 = u^2 + 2(9.81)(-6.8)$$

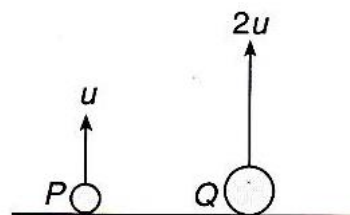
$$u = 11.5506 \text{ m s}^{-1}$$

$$\approx 11.6 \text{ m s}^{-1}$$

D

18. As shown in the above figure, two objects, P and W, are projected at initial speeds u and $2u$ respectively. Which one has a higher maximum height and which one returns earlier?

	Higher max. height	Returns earlier
A.	P	P
B.	Q	Q
C.	Q	P
D.	Q	both



C

19. A ball on the ground is projected vertically upwards at time $t = 0$ with an initial velocity of 12 m s^{-1} . Neglecting air resistance, which of the following statements is/are correct?

- (1) The maximum height reached by the ball is 7.34 m .
 (2) The displacement of the ball at $t = 2.4 \text{ s}$ is 14.4 m .
 (3) The average velocity of the ball from $t = 0$ to $t = 2.4 \text{ s}$ is 6 m s^{-1} .

- A. (1) only
 B. (2) only
 C. (1) and (3) only
 D. (2) and (3) only

$$\begin{aligned} (1) \quad 0 &= 12^2 - 2(9.81)(S) \quad S = 7.34\text{m} \\ (2) \quad s &= 12(2.4) - \frac{1}{2}(9.81)2.4^2 = 0.5472\text{m} \\ (3) \quad v &= \frac{0.5472}{2.4} = 0.228\text{s} \end{aligned}$$

A

20. Two small identical objects P and Q are released from the top of a building 80 m above the ground. Q is released 1 s after P. Neglecting air resistance, what is the maximum vertical separation between P and Q in the air?

- A. 5 m
 B. 10 m
 C. 35 m
 D. 45 m

For simplicity, take g to be 10 m s^{-2} .
 For P, $s_P = ut + \frac{1}{2}at^2 = \frac{1}{2}(10)t^2 = 5t^2$
 For Q, $s_Q = 5(t-1)^2$
 $\Delta s = 5t^2 - 5(t-1)^2 = 10t - 5$
 Maximum separation occurs when P just reaches the ground.
 $\therefore (80) = 5t^2 \quad \therefore t = 4 \text{ s}$
 $\therefore \Delta s_{\text{max}} = 10 \times 4 - 5 = 35 \text{ m}$

C

21. A ball is thrown upwards from the bottom of building. It can just reach the top of a building and takes 2.5 s . What is the time required for the ball to fall back to the middle of the building from the top?

- A. 1.33 s
 B. 1.55 s
 C. 1.77 s
 D. 1.99 s

In upward journey, ball move to top of building within 2.5s . By symmetry, it takes the same time for downward journey to the bottom of building ($x \text{ m}$).

$$x = \frac{1}{2}g(2.5)^2$$

For case when it fall back to the middle of building ($\frac{x}{2} \text{ m}$),

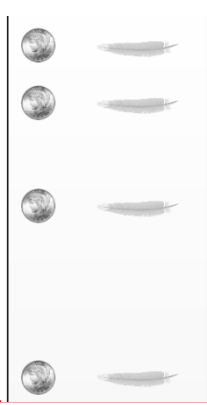
$$\begin{aligned} \frac{1}{2}x &= \frac{1}{2}gt^2 \\ \frac{1}{2}\left[\frac{1}{2}g(2.5)^2\right] &= \frac{1}{2}gt^2 \\ t &= 1.77 \text{ s} \end{aligned}$$

C

22. A coin and a feather are released from the top of a vacuum tube at the same time. The positions of the coin at three consecutive seconds are shown below.

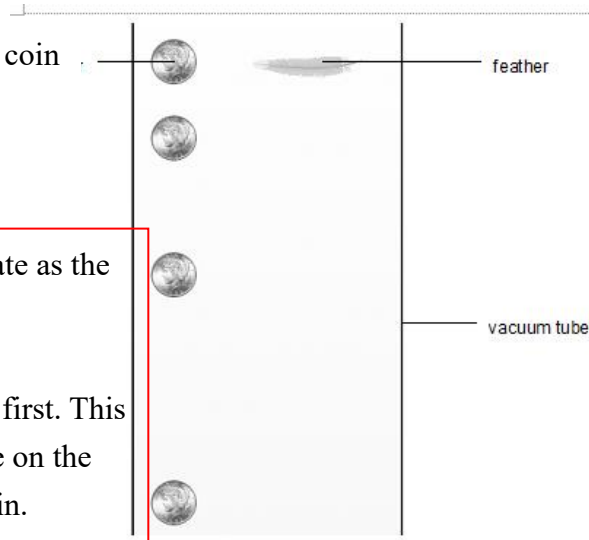
- (a) (i) Draw, on the above diagram, the corresponding positions of the feather.
(ii) Compare the motion of the feather with that of the coin.
(b) Suppose the tube is filled with air. Which object, the coin or the feather, will reach the bottom of the tube first? Why?

(a) (i)



(a)(ii) The feather falls at the same rate as the coin.

(b) The coin will reach the bottom first. This is because the effect of air resistance on the feather is greater than that on the coin.



23. In the figure as shown, John releases ball *A* at time $t = 0$ from rest at a position 7 m above the ground. At the same time, ball *B* is projected vertically upwards from the ground with an initial speed of 8 m s^{-1} .

- (a) Find the time required by ball *B* to reach its maximum height. (2 marks)
(b) When and at what height will the balls meet in air? (3 marks)

Take the upward direction as positive.

- (a) Applying $v = u + gt$, the time required by ball *B* to reach its maximum height is

$$t = \frac{v - u}{g}$$

$$= \frac{0 - 8}{-9.81}$$

$$= 0.815 \text{ s}$$

(1M+1A)

- (b) The displacement of ball *A* from the ground is $7 + \frac{1}{2}gt^2$.

(1M)

The displacement of ball *B* from the ground is $8t + \frac{1}{2}gt^2$.

(1M)

When the balls meet in air,

$$7 + \frac{1}{2}gt^2 = 8t + \frac{1}{2}gt^2$$

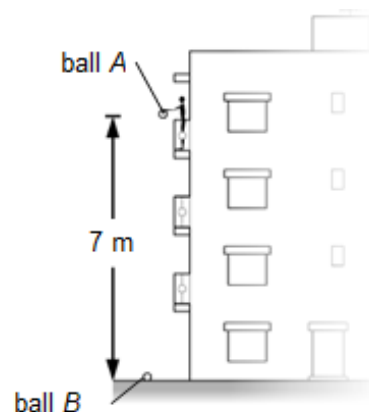
$$t = 0.875 \text{ s}$$

The balls meet 0.875 s after ball *A* is released. At the same instant, the

displacement of the balls above the ground is a $7 + \frac{1}{2} \times (-9.81) \times 0.875^2$

$$= 3.24 \text{ m}$$

(1A)



24. A stone is projected vertically up with a speed of 10 m s^{-1} from a cliff which is 20 m above the sea level. Calculate
- the maximum height of the stone above the sea level
 - the time for the stone to reach the bottom of the cliff
 - the speed of the stone just before it hits the water surface
- Take $g = 9.81 \text{ m s}^{-2}$ and neglect air resistance

- (a) Let h be the maximum height of the stone above the cliff.

By $v^2 = u^2 + 2as$, taking downward positive,

$$0 = 10^2 + 2g \cdot (-h)$$

$$h = 5.1 \text{ m}$$

The maximum height above the sea level is

$$20 + 5.1 \text{ m} = 25.1 \text{ m}$$

- (b) By $s = ut + \frac{1}{2}at^2$, taking downward positive.

$$20 = -10t + \frac{1}{2} \times (9.81)t^2$$

$$t = 3.28 \text{ or } -1.24 \text{ s (rejected)}$$

$\therefore$ time required is 3.28 s

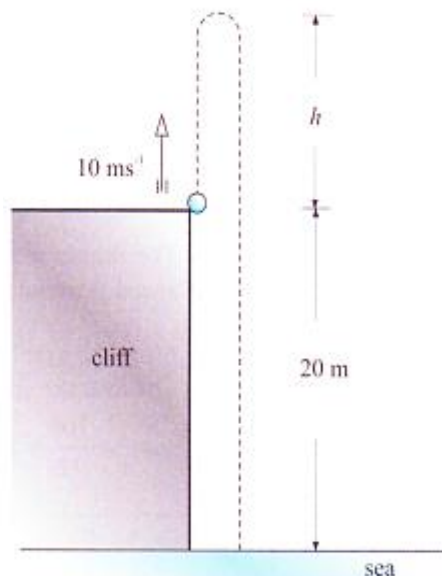
- (c)

$$v^2 - u^2 = 2as$$

$$v^2 - (-10)^2 = 2 \times 9.81 \times 20$$

$$\therefore v = 22.2 \text{ m s}^{-1}$$

The speed of the stone just before hit the water surface is 22.2 m s^{-1}



25. At time $t = 0$, Simon throws a tennis ball vertically upwards at 7 m s^{-1} from a height of 1.70 m above the ground. When the ball falls back to a height of 3.30 m , Simon hits it. Neglect air resistance and take $g = 9.81 \text{ m s}^{-2}$.

- (a) What is the maximum height above the ground reached by the ball?
(b) When does Simon hit the ball?



- (a) Take upwards as positive.

Consider the upward motion to the maximum height.

$$u = +7 \text{ m s}^{-1}, v = 0, a = -9.81 \text{ m s}^{-2}, s = ?$$

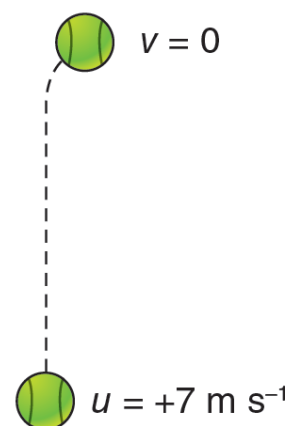
$$\text{By } v^2 = u^2 + 2as,$$

$$0^2 = 7^2 + 2 \times (-9.81) \times s$$

$$s = 2.50 \text{ m}$$

Maximum height above the ground

$$= 1.70 + 2.50 = 4.20 \text{ m}$$



- (b) $u = +7 \text{ m s}^{-1}, a = -9.81 \text{ m s}^{-2},$

$$s = +(3.30 - 1.70) = +1.60 \text{ m}, \quad t = ?$$

$$\text{By } s = ut + \frac{1}{2} at^2,$$

$$1.60 = 7t + \frac{1}{2} \times (-9.81) \times t^2$$

$$t = 1.14 \text{ s} \quad \text{or} \quad 0.286 \text{ s (rejected)}$$

Simon hits the ball at $t = 1.14 \text{ s}$.

